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SCHOOL of: Business

ASSIGNMENT COVER SHEET

STUDENT DETAILS

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SUBJECT AND TUTORIAL DETAILS

Subject Name: Programming for Data Science **Subject code:** COMP7024

Tutorial Group: Click or tap here to enter text. **Day:** Wednesday **Time:** 1800

Lecturer or Tutor name: Vishal Patel

ASSIGNMENT DETAILS

Title: Assignment **Length:** Click or tap here to enter text.

Due Date: 13/03/2026 **Date submitted: 13/03/2026**

Home campus: Parramatta City

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COMP7024 Assignment

Zarrar Khan (22168452)

2026-03-13

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#template to answer the questions

Q 1

Explanation

This analysis aims at analyzing the performance of products in terms of revenue. The Orders data set will be combined with the Product Info data set with the common variable of StockCode in order to conduct this analysis. This union enables every transaction in the Orders table to have the corresponding product description to enable the identification of products in the analysis.

As the dataset does not have a revenue variable, the revenue per transaction is calculated in the following formula: Revenue = Quantity x Unitprice. This is the value of a monetary amount per individual transaction. The transaction-level revenue is then joined to the product level adding the data by StockCode and Description and adding the values of the revenue to obtain the TotalRevenue of each product.

The products are arranged in descending order starting with the highest revenue to the lowest revenue after being calculated as the total revenue of each product in order to know the most performing products and those least performing products. Based on such ranking, the best 10 and worst 10 products are obtained to compare between high and low revenue products. Besides this, the total revenue of the Top 3 products is counted and compared to the total revenue as a percentage to assess the degree of revenue concentration among the most performing products.

In order to further break down on the issue of revenue concentration, the products are separated into two groups namely the Top 10 products and all the other products. The contribution of each group to total revenue is compared and the visualisation of the revenue generated by each group is made.

The analysis is based on the assumption that the revenue is accurately represented by the product of quantity and unit price, and every StockCode is associated with the appropriate product description. Any gaps in

terms of revenue computations are disregarded so as to make sure that they do not influence the consolidated outcomes.

Code

```
#Write your code here
# Read the Orders dataset (transactions)
orders = read_csv("project_orders.csv")

# Read the Product Info dataset (Description)
products = read_csv("project_product_info.csv")

# Join Product info table in project orders table
# Sales would be the new dataset without disrupting the orders dataset
sales = orders %>%
  left_join(products, by = "StockCode") # attach Description to each order line

# Creating a new column "Revenue" through mutate
sales = sales %>%
  mutate(Revenue = Quantity * UnitPrice)
# Calculating Total Revenue Per Product
product_rev = sales %>%
  group_by(StockCode, Description) %>% # Grouping the products by their description
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE), # sums up all the revenue for each product
            .groups = "drop") %>% # Remove the grouping after summaries
  arrange(desc(TotalRevenue))

top10 = product_rev %>% slice_head(n = 10) # first 10 rows (highest revenue)
bottom10 = product_rev %>% slice_tail(n = 10) # last 10 rows (lowest revenue)


total_rev_all = sum(product_rev$TotalRevenue, na.rm = TRUE)  # total revenue across all products making

top3_summary = product_rev %>%
  slice_head(n = 3) %>%   # take top 3 products
  summarise(Top3Revenue = sum(TotalRevenue, na.rm = TRUE)) %>%  # sum their revenue an make only one ro
  mutate(Top3Percent = 100 * Top3Revenue / total_rev_all)        # % of total

# Display in Table format
top10 %>%
  kbl(caption = "Q1: Top 10 Products by Total Revenue") %>% # Adds caption for the table
kable_styling(full_width = FALSE) # Compacting the width
```

```
bottom10 %>%
kbl(caption = "Q1: Bottom 10 Products by Total Revenue") %>%
kable_styling(full_width = FALSE)
```

```
top3_summary %>%
  kbl(caption = "Q1: Top 3 Products Revenue Contribution") %>%
  kable_styling(full_width = FALSE)
```

Table 1: Q1: Top 10 Products by Total Revenue

StockCode	Description	TotalRevenue
DOT	DOTCOM POSTAGE	206245.48
22423	REGENCY CAKESTAND 3 TIER	164762.19
47566	PARTY BUNTING	98302.98
85123A	WHITE HANGING HEART T-LIGHT HOLDER	97894.50
85099B	JUMBO BAG RED RETROSPOT	92356.03
23084	RABBIT NIGHT LIGHT	66756.59
POST	POSTAGE	66230.64
22086	PAPER CHAIN KIT 50'S CHRISTMAS	63791.94
84879	ASSORTED COLOUR BIRD ORNAMENT	58959.73
79321	CHILLI LIGHTS	53768.06

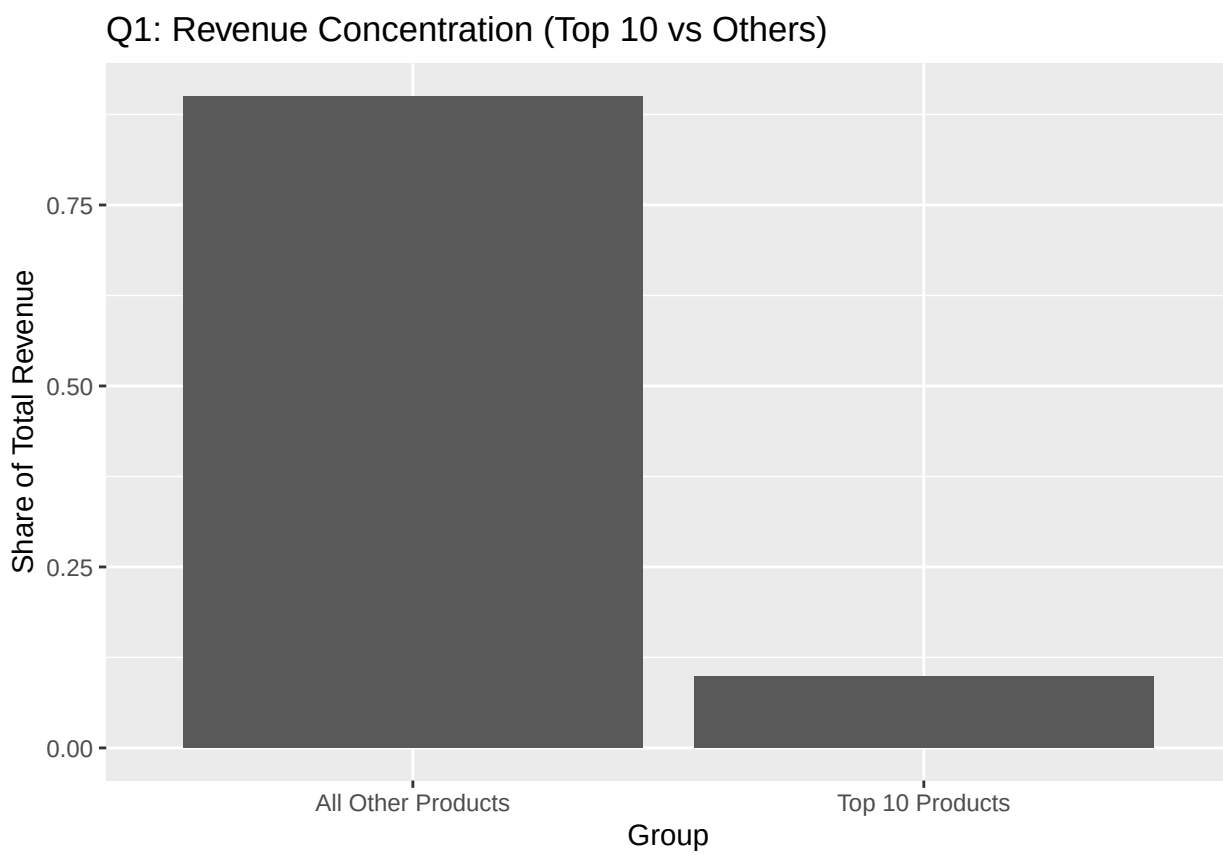
Table 2: Q1: Bottom 10 Products by Total Revenue

StockCode	Description	TotalRevenue
35400	WOODEN BOX ADVENT CALENDAR	-45.700
85063	CREAM SWEETHEART MAGAZINE RACK	-46.850
79323W	WHITE CHERRY LIGHTS	-54.000
S	SAMPLES	-3049.390
D	Discount	-5696.220
BANK CHARGES	Bank Charges	-7175.639
CRUK	CRUK Commission	-7933.430
B	Adjust bad debt	-11062.060
M	Manual	-68674.190
AMAZONFEE	AMAZON FEE	-221520.500

Table 3: Q1: Top 3 Products Revenue Contribution

Top3Revenue	Top3Percent
469310.7	4.814555

```r
# Making plots

rev_concentration <- product_rev %>%
  mutate(rank = row_number(),
         group = if_else(rank <= 10, "Top 10 Products", "All Other Products")) %>% # Creating 2 columns
  # if_else(Condition, If true, if false)
  group_by(group) %>%
  summarise(group_revenue = sum(TotalRevenue), .groups = "drop") %>%
  mutate(share = group_revenue / sum(group_revenue))

ggplot(rev_concentration, aes(x = group, y = share)) +
  geom_col() +
  labs(
    title = "Q1: Revenue Concentration (Top 10 vs Others)",
    x = "Group",
    y = "Share of Total Revenue"
  )
```



Interpretation

The outcome of the revenue analysis indicates that few products produce a large percentage of the revenue in the company. The table of the 10 products with highest revenue indicates that the products with the highest revenue of all the products in the data are: DOTCOM POSTAGE, REGENCY CAKE STAND 3 TIER and PARTY BUNTing. Specifically, DOTCOM POSTAGE generates the largest revenue of 206,245.48, then

there is REGENCY CAKE STAND 3 TIER with 164,762.19. These products seem to be the main sources of total business revenue and probably high demand product or a commonly bought product.

Conversely, the Bottom 10 products based on total revenue have a number of values that are negative revenue like AMAZON FEE, Manual, Adjust bad debt and Discount. The existence of some negative values indicates that these entries are financial adjustments, transaction fees, or correction of some service but not the sale of products. As an illustration, AMAZON FEE has the lowest [?]221,520.50 that is probably not revenue earned as a result of selling goods and services, but rather transaction fees or other costs associated with using the platform.

The revenue contribution of the Top 3 products revealed that the three highest revenue products in terms of contribution make up a revenue amount of about 469,310.7 and this is equivalent to about 4.81 percent of the total product revenue. This means that though these products bring about highest individual revenues, the total revenue is spread over a vast variety of different products.

The revenue share visualisation also supports this observation. The graph shows that the 10 products which are in the top sellers contribute about 10 percent of the total revenue with the other products taking up about 90 percent of the revenue. This implies that the company does not have a small number of products that it relies on to generate its revenue but rather has a wide product catalogue.

This trend, as far as business is concerned, signifies an eclectic revenue composition, in which numerous products contribute small portions of the operating sales compared to the consequences of a limited quantity of products. Diversification may be advantageous since it decreases the reliance on single items and allows staying on the level of income even in case of the decrease in the demand of some of the products. Simultaneously, the recognition of the best performing products is useful in terms of inventory control, advertising plans, and promotional activities since these products might be some of the primary customer movements.

Q 2

Explanation

This analysis aims at reviewing the performance of the products in terms of revenue. In order to conduct this analysis, Orders dataset is first joined with Product Info dataset using the common variable of StockCode. This join means that every transaction in the Orders data set will incorporate the description of the product in question so that during the analysis products will be easily identified.

As the dataset does not give a revenue variable, revenue is computed on a transaction-by-transaction basis based on the following formula Revenue = Quantity x Unitprice. This value is in the form of monetary value of one transaction. The transaction-level revenue is further aggregated into the product level by combining the data by StockCode and Description and adding the values of revenue to get the TotalRevenue of each product. This analysis proposal aims at the identification of high-value customers on the basis of total spending and the analysis of their purchasing behaviour. The missing values of CustomerID in some of the transactions in the dataset are not included in the analysis since missing customerID can represent several unidentifiable customers as opposed to one identifiable person. The deletion of such records will make sure the analysis considers customers that can be singly identified.

The final spending per customer is computed, after cleaning up the dataset, by summing the Revenue generated on all the transactions of a customer, in terms of CustomerID. The revenue variable used to be expressed as Quantity x UnitPrice which is the amount of money paid as a result of every transaction. The customers are then ranked according to a total spending list to end up with the Top 5 highest spending customers.

To examine the buying behaviour of such high-value customers, the data is narrowed down to the Top 5 customers. The data is further aggregated on CustomerID, StockCode and Description and the total quantity that has been purchased of a particular product is summed up. This enables the determination of products that were most frequently bought by the high value customers. To be more precise, a visualisation is retained with only the Top 5 most purchased products per customer.

This analysis makes the assumption that the revenue computation is reflecting the accurate customer spending and that every non-missing CustomerID is the unique customer. So it is assumed that the product descriptions are accurate in relation to stock codes and that the overall amount purchased is a relevant measure of any product preference. Once the total revenue of each product is computed, the products would be ranked highest to lowest in terms of revenue in order to find the best and the worst performing products. Out of this rank, the Top 10 and Bottom 10 products are selected to make comparisons between high and low revenue generating products. Moreover, the sum of top 3 products revenue is obtained and converted into percentage of overall revenue to assess the degree of revenue concentration of the most successful products.

In order to further breakdown the revenue concentration, the products are classified into two, namely the Top 10 products and all other products. Visualisation is done by calculating the revenue generated by each group and comparing the contribution made to the total revenue.

The analysis also makes some assumptions regarding revenue, which could be appropriately represented by the product description of the quantity times unit price, and the fact that every StockCode is matched to a specific product description. Any gaps in the revenue computation are ignored so as to make sure that it does not influence the aggregated outcomes.

Code

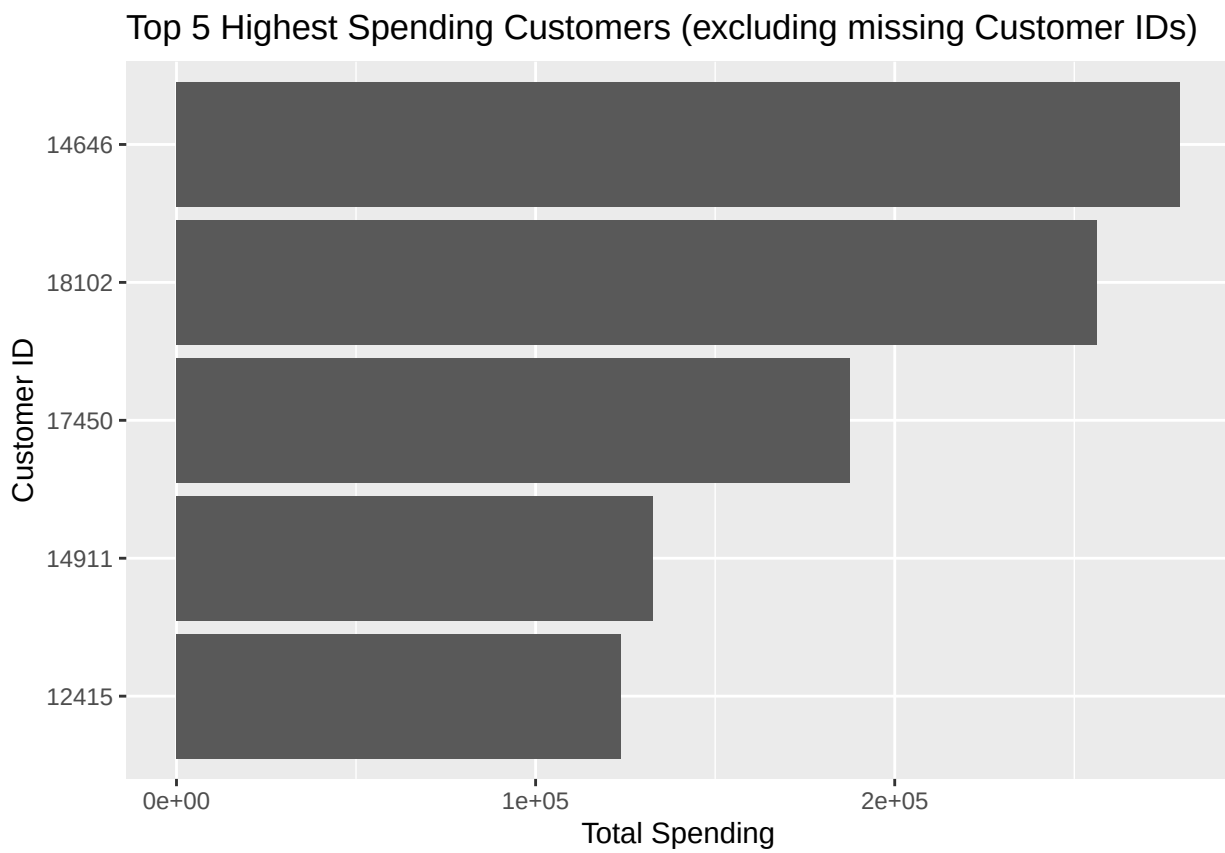
```r
#Write your code here
# Calculating total spending for each customer
# Remove rows where CustomerID is missing
# Missing CustomerID values are excluded because NA may represent multiple unidentified customers rathe
sales_customers = sales %>%
  filter(!is.na(CustomerID)) # !is.na only selects those entries which are NOT missing

# Calculate total spending for each customer
# We group by CustomerID and sum Revenue
customer_spending = sales_customers %>%
  group_by(CustomerID) %>%
  summarise(TotalSpending = sum(Revenue, na.rm = TRUE),
            .groups = "drop") %>%
  arrange(desc(TotalSpending))   # highest spender first

# ------------------------------------------------------------
# Extract the Top 5 highest spending customers
# Since the data is sorted in descending order, the first 5 rows are the top 5 customers

top5_customers = customer_spending %>%
  slice_head(n = 5)

# Display Top 5 customers
top5_customers
```

```
## # A tibble: 5 x 2
##   CustomerID TotalSpending
##        <dbl>         <dbl>
## 1      14646       279489.
## 2      18102       256438.
## 3      17450       187482.
## 4      14911       132573.
## 5      12415       123725.
```

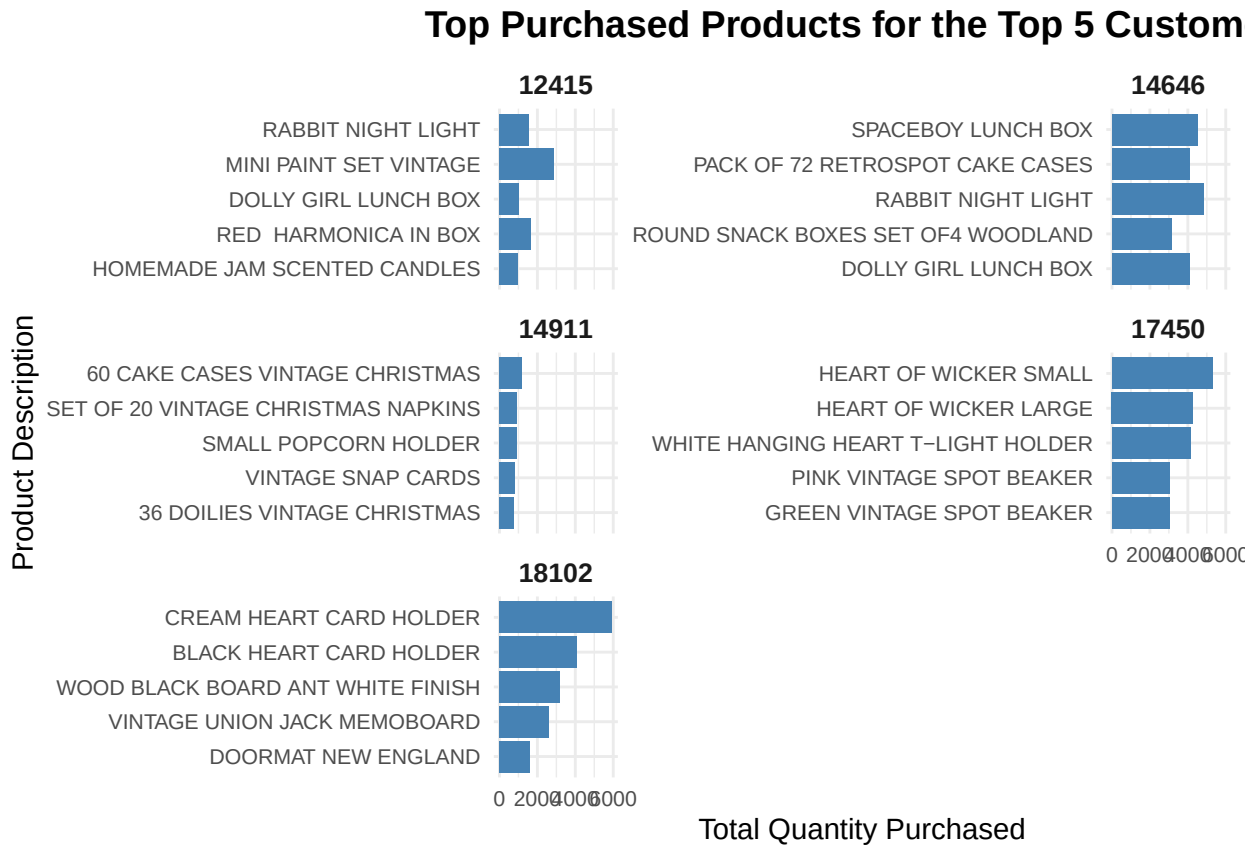
```r
# Store the CustomerIDs of the Top 5 customers
top5_ids = top5_customers$CustomerID

# Analyse purchasing behaviour of the Top 5 customers
# Keep only transactions from these Top 5 customers
# Then group by customer and product
# Then sum Quantity to find which products they purchased the most
top_customer_products = sales_customers %>%
  filter(CustomerID %in% top5_ids) %>% # %in% means that CustomerID which are available in top5_ids
  group_by(CustomerID, StockCode, Description) %>%
  summarise(TotalQuantity = sum(Quantity, na.rm = TRUE),
            .groups = "drop") %>%
  arrange(CustomerID, desc(TotalQuantity))

# ------------------------------------------------------------
# Keep only the top 5 most purchased products for each of the Top 5 customers
top_customer_products_top5 = top_customer_products %>%
  group_by(CustomerID) %>%
  slice_head(n = 5) %>%
  ungroup()

# Display top products for each customer
top_customer_products_top5
```

```
## # A tibble: 25 x 4
##    CustomerID StockCode Description                        TotalQuantity
##         <dbl> <chr>     <chr>                                    <dbl>
##  1      12415 22492     MINI PAINT SET VINTAGE                    2880
##  2      12415 21915     RED  HARMONICA IN BOX                     1680
##  3      12415 23084     RABBIT NIGHT LIGHT                        1536
##  4      12415 22630     DOLLY GIRL LUNCH BOX                      1024
##  5      12415 22969     HOMEMADE JAM SCENTED CANDLES               970
##  6      14646 23084     RABBIT NIGHT LIGHT                        4801
##  7      14646 22629     SPACEBOY LUNCH BOX                        4492
##  8      14646 21212     PACK OF 72 RETROSPOT CAKE CASES           4104
##  9      14646 22630     DOLLY GIRL LUNCH BOX                      4096
## 10      14646 22326     ROUND SNACK BOXES SET OF4 WOODLAND        3120
## # i 15 more rows
```

```r
# Bar chart of the Top 5 highest spending customers
# as.factor keeps the customerID as a category rather than a numeric value
# reorder makes the graph in a specific direction
ggplot(top5_customers, aes(x = reorder(as.factor(CustomerID), TotalSpending), y = TotalSpending)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 5 Highest Spending Customers (excluding missing Customer IDs)",
    x = "Customer ID",
    y = "Total Spending"
  )
```




```r
# Faceted bar chart showing the top 5 most purchased products
# for each of the Top 5 customers
ggplot(top_customer_products_top5,
       aes(x = reorder(Description, TotalQuantity), y = TotalQuantity)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  facet_wrap(~ CustomerID, scales = "free_y", ncol = 2) +
  labs(
    title = "Top Purchased Products for the Top 5 Customers",
    x = "Product Description",
    y = "Total Quantity Purchased"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    axis.text.y = element_text(size = 8),
    axis.text.x = element_text(size = 8),
    axis.title = element_text(size = 11),
    strip.text = element_text(size = 10, face = "bold")
  )
```



Interpretation

As it can be seen, the Top 5 highest spending customers will be CustomerID 14646, 18102, 17450, 14911 and 12415 with the total spending amounts of about 279,489, 256,438, 187,482, 132,572 and 123,725 respectively. The customer with the highest spending is Customer14646 though the gap between the top and the second highest spending customers is not as high as in the previous analysis whereby missing customer IDs were counted.

The purchasing behaviour analysis will unravel the product preferences of these high value customers. As an example, the top spending customer Customer 14646 bought large quantities of products like RABBIT NIGHT LIGHT, SPACEBOY LUNCH BOX, and PACK OF 72 RETROspot CAKE CASES, which means that the decorative household and gift products are in high demand. On the same note, Customer 18102 bought large volumes of the products like CREAM HEART CARD HOLDER and BLACK HEART CARD HOLDER and this may indicate a taste towards decorative accessories.

Purchasing patterns are also different among other customers. As an example, Customer 17450 regularly bought products of the types HEART OF WICKER SMALL and HEART of WICKER LARGE, and Customer 12415 often bought such products as MINI PAINT SET VINTAGE and RED HARMONICA IN Box. These trends indicate that the high-value customers tend to make large volumes of particular types of products instead of having a broad range of products.

In business terms, understanding of product preferences and identifying high-value customers will serve the purpose of managing relationships with customers and marketing to specific people. The information can be used by businesses to create personalised promotions, approval of similar products, and keeping of popular products at good stock levels. It is especially important to retain these high-value customers as they generate a large percentage of the total income of the company.

Q 3

Explanation

This analysis aims at analyzing the consumption trends of goods in various countries and comparing the manner in which the revenue differs among the world markets. To obtain this, the dataset will be clustered on Country and the amount of revenue earned in each country will be determined by adding up the Revenue variable in all transactions made. The revenue variable has been determined in the past as Quantity x UnitPrice, which is the financial worth of a single purchase.

To determine the consumption trend of the product in each country the dataset is further classified by Country, Stockcode and Description. The amount sold in each country of every product is then determined. This will enable in determination of a single product that records the highest level of sale in the respective countries which will be the popular product in the respective market.

The country-level revenue statistics and the most bought product in the country are then summed up into one in form of a summary table. The following table gives a summary of the performance of revenue and also demand of products in various countries. One can also visualise the revenue earned per country by a bar chart and it will be easy to compare the performance of each market.

The analysis presupposes that the computation of the revenues (Quantity x UnitPrice) is the true value of every transaction. The Country variable is also assumed to be accurate in identifying the location of the transaction and product description to match stock codes that it is associated with. Also, the amount of goods sold is believed to be a proper indicator to determine the most favored product in each country.

Code

```r
#Write your code here

# Calculate total revenue for each country
# Group the dataset by Country and sum the Revenue
# ------------------------------------------------------------
country_revenue = sales %>%
  group_by(Country) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE),
            .groups = "drop") %>%
  arrange(desc(TotalRevenue))

# Display revenue by country
country_revenue
```

```
## # A tibble: 38 x 2
##    Country        TotalRevenue
##    <chr>                 <dbl>
##  1 United Kingdom     8187806.
##  2 Netherlands         284662.
##  3 EIRE                263277.
##  4 Germany             221698.
##  5 France              197404.
##  6 Australia           137077.
##  7 Switzerland          56385.
##  8 Spain                54775.
##  9 Belgium              40911.
```

```
## 10 Sweden               36596.
## # i 28 more rows
```

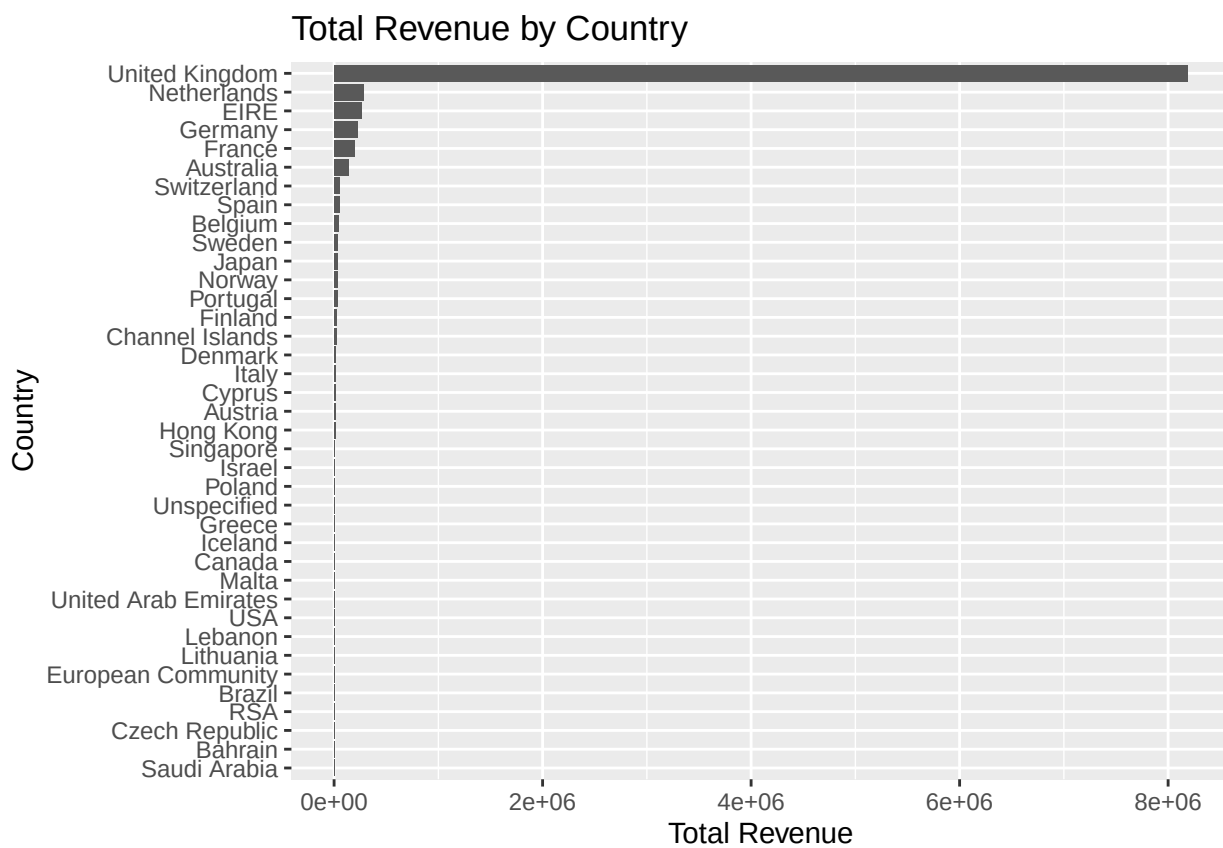
```r
# Calculating total quantity sold for each product within each country
country_product_quantity = sales %>%
  group_by(Country, StockCode, Description) %>%
  summarise(TotalQuantity = sum(Quantity, na.rm = TRUE),
            .groups = "drop")


# Identify the most purchased product in each country based on highest quantity
top_product_country = country_product_quantity %>%
  group_by(Country) %>%
  slice_max(TotalQuantity, n = 1) %>% # For each country keep the row with the highest TotalQuantity
  ungroup()

top_product_country
```

```
## # A tibble: 62 x 4
##    Country   StockCode Description                        TotalQuantity
##    <chr>     <chr>     <chr>                                    <dbl>
##  1 Australia 22492     MINI PAINT SET VINTAGE                    2916
##  2 Austria   21918     SET 12 KIDS COLOUR  CHALK STICKS           288
##  3 Bahrain   23076     ICE CREAM SUNDAE LIP GLOSS                  96
##  4 Belgium   21212     PACK OF 72 RETROSPOT CAKE CASES            480
##  5 Brazil    21430     SET/3 RED GINGHAM ROSE STORAGE BOX          24
##  6 Brazil    22630     DOLLY GIRL LUNCH BOX                        24
##  7 Brazil    22697     GREEN REGENCY TEACUP AND SAUCER             24
##  8 Brazil    22698     PINK REGENCY TEACUP AND SAUCER              24
##  9 Brazil    22699     ROSES REGENCY TEACUP AND SAUCER             24
## 10 Brazil    22722     SET OF 6 SPICE TINS PANTRY DESIGN           24
## # i 52 more rows
```

```r
# Combine country revenue with the top product information into one summary table

country_summary = country_revenue %>%
  left_join(top_product_country, by = "Country") # joining country_revenue and top_product_country tabl

country_summary
```

```
## # A tibble: 62 x 5
##    Country        TotalRevenue StockCode Description               TotalQuantity
##    <chr>                 <dbl> <chr>     <chr>                           <dbl>
##  1 United Kingdom     8187806. 22197     SMALL POPCORN HOLDER            52928
##  2 Netherlands         284662. 23084     RABBIT NIGHT LIGHT               4801
##  3 EIRE                263277. 22197     SMALL POPCORN HOLDER             1809
##  4 Germany             221698. 22326     ROUND SNACK BOXES SET OF~        1218
##  5 France              197404. 23084     RABBIT NIGHT LIGHT               4023
##  6 Australia           137077. 22492     MINI PAINT SET VINTAGE           2916
##  7 Switzerland          56385. 22554     PLASTERS IN TIN WOODLAND~         639
##  8 Spain                54775. 84997D    PINK 3 PIECE POLKADOT CU~        1089
##  9 Belgium              40911. 21212     PACK OF 72 RETROSPOT CAK~         480
## 10 Sweden               36596. 22492     MINI PAINT SET VINTAGE           2916
## # i 52 more rows
```

```r
# Visualise revenue by country
ggplot(country_revenue,
       aes(x = reorder(Country, TotalRevenue), y = TotalRevenue)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Total Revenue by Country",
    x = "Country",
    y = "Total Revenue"
  )
```



Interpretation

The findings reveal that there exist high disparities in revenue generation among different countries. The United Kingdom is the top revenue producer by far with a total revenue of about 8.19 million which is significantly higher than the rest of the countries in the data set. This means that the United Kingdom is the main market of the business and that it makes the major contribution to the overall sales.

The countries that follow in terms of revenue generation are Netherlands, EIRE, Germany and France, albeit at a very low level when compared to the level in the United Kingdom. This implies that the business is international but the majority of transactions are made in the UK market. Other nations like Australia, Switzerland, Spain and Belgium are sources of moderate revenue levels which means that there are smaller yet significant markets internationally.

The analysis of the product consumption indicates that some of the products seem to reoccur as the most bought products in several countries. Also, the most popular product bought in United Kingdom and EIRE is

SMALL POPCORN HOLDER and in several countries such as Netherlands, France and Japan it is a product called RABBIT NIGHT LIGHT. On the same note, some products like MINI PAINT SET VINTAGE are featured among the popular products in Australia and Sweden. This implies that there are products which are widely accepted in the international market and are always at demand in various markets.

On the business side, such findings reveal apparent regional variations in buying behaviour. The high level of dominance by the United Kingdom implies that the core customers of the company are found within the United Kingdom. Nevertheless, the fact that the popular items are available in various countries suggests the possibility of going international. This information can be used by business to target marketing and inventory planning towards products that are in high demand in a particular region and also seek opportunities to increase their presence in emerging international markets.

Q 4

Explanation

This analysis aims to review the the changes in sales performance through the periods of time and also to find trends in monthly sales revenue and product demand. In order to conduct this analysis, firstly, the InvoiceDate variable was changed to date-time format to allow extracting information related to time. Based on this variable, the year and month of each transaction were derived and put together as a new variable (YearMonth) which is used as a point of time in which the transaction took place.

This time variable was used to aggregate the transactions by the YearMonth and determine the total revenue made in the month. This was achieved through addition of the revenue values of all the transactions made in each month. Such aggregation made it possible to identify months in which sales are the greatest and analyze how the revenue varies over time.

To explore the product performance with time, the dataset has been classified into months, the product code, and product description and subsequently the total number of products sold in a month of each product was computed. The products that sold the least amount of products in each month were then established. Following the identification of these low-performing products in every month, the analysis of the analysis tallyed the number of months that each product featured in the list of the lowest selling products. This resulted in the identification of products that are always doing poorly over a number of months.

The analysis is based on the assumption that the calculation of the revenues (Quantity x UnitPrice) is the true measure of the value of every transaction. It also makes assumptions that the InvoiceDate variable adequately represents the point in time the transactions took place and that the classification of the data based on the month gives a relevant degree of aggregation in studying the trend of sales over time. Moreover, it is assumed that low selling products are items characterized by the constant low customer demand in relation to other items in the catalogue.

Code

```
#Write your code here
# Convert InvoiceDate and extract Year and Month
sales_time = sales %>%
  mutate(
    InvoiceDate = as.POSIXct(InvoiceDate), # Converts it into proper date
    YearMonth = as.Date(paste0(format(InvoiceDate,"%Y-%m"),"-01")))

# Calculate total revenue for each month
monthly_revenue = sales_time %>%
  group_by(YearMonth) %>%
```

```r
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop") %>%
  arrange(YearMonth)

monthly_revenue
```

```
## # A tibble: 298 x 2
##    YearMonth  TotalRevenue
##    <date>            <dbl>
##  1 0001-02-01       28433.
##  2 0001-03-01       25472.
##  3 0001-04-01       24392.
##  4 0001-05-01        6965.
##  5 0001-06-01       20191.
##  6 0001-07-01       13172.
##  7 0001-08-01       21363.
##  8 0001-09-01       37297.
##  9 0001-11-01       28742.
## 10 0001-12-01      110047.
## # i 288 more rows
```

```r
# Identify months with highest sales
top_months = monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5) # Shows only top 5 months

top_months
```

```
## # A tibble: 5 x 2
##   YearMonth  TotalRevenue
##   <date>            <dbl>
## 1 0008-12-01      125608.
## 2 0007-12-01      120135.
## 3 0014-11-01      112141.
## 4 0001-12-01      110047.
## 5 0020-09-01      109286.
```

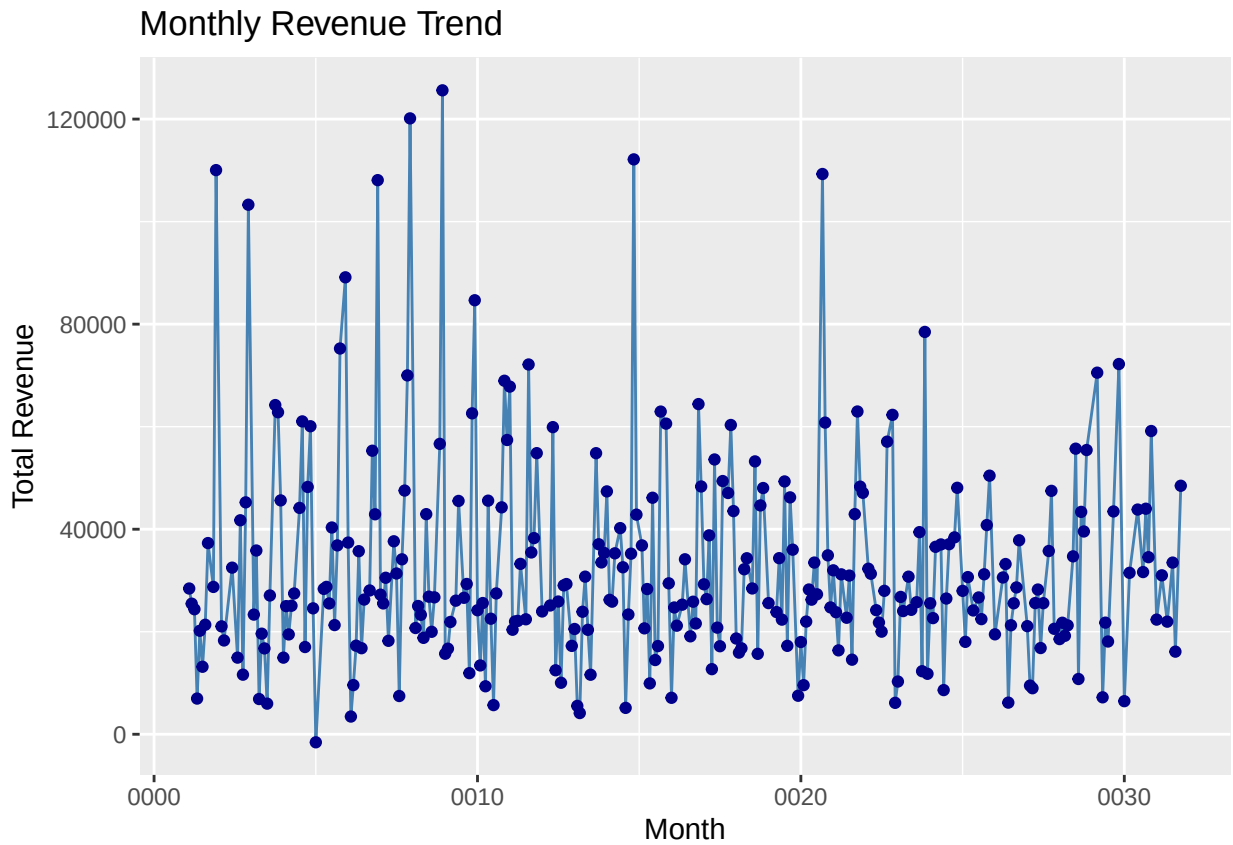
```r
# Analyse product performance over time
# Calculate quantity sold per product each month
product_month_sales = sales_time %>%
  group_by(YearMonth, StockCode, Description) %>%
  summarise(
    TotalQuantity = sum(Quantity, na.rm = TRUE),
    .groups = "drop")


# Identify lowest selling products each month
lowest_products_month = product_month_sales %>%
  group_by(YearMonth) %>%
  slice_min(TotalQuantity, n = 5) %>%
  ungroup()

lowest_products_month
```

```
## # A tibble: 8,022 x 4
##    YearMonth  StockCode Description                        TotalQuantity
##    <date>     <chr>     <chr>                                    <dbl>
##  1 0001-02-01 47591B    SCOTTIES CHILDRENS APRON                  -206
##  2 0001-02-01 22111     SCOTTIE DOG HOT WATER BOTTLE               -72
##  3 0001-02-01 84748     FOLK FELT HANGING MULTICOL GARLAND         -52
##  4 0001-02-01 21448     12 DAISY PEGS IN WOOD BOX                  -40
##  5 0001-02-01 22162     HEART GARLAND RUSTIC PADDED                -37
##  6 0001-03-01 22797     CHEST OF DRAWERS GINGHAM HEART             -14
##  7 0001-03-01 20793     BLUE TILED TRAY                             -5
##  8 0001-03-01 20618     QUEEN OF THE SKIES PASSPORT COVER            1
##  9 0001-03-01 20658     RED RETROSPOT LUGGAGE TAG                    1
## 10 0001-03-01 20761     BLUE PAISLEY SKETCHBOOK                      1
## # i 8,012 more rows
```

```r
# Identify products that repeatedly appear among lowest sellers across months
consistent_low_products = lowest_products_month %>%
  group_by(StockCode, Description) %>%
  summarise(MonthsAsLowSeller = n(),.groups = "drop") %>%
  arrange(desc(MonthsAsLowSeller))

consistent_low_products
```

```
## # A tibble: 2,736 x 3
##    StockCode Description                     MonthsAsLowSeller
##    <chr>     <chr>                                       <int>
##  1 M         Manual                                         35
##  2 22424     ENAMEL BREAD BIN CREAM                         15
##  3 22839     3 TIER CAKE TIN GREEN AND CREAM                14
##  4 21539     RED RETROSPOT BUTTER DISH                      13
##  5 22180     RETROSPOT LAMP                                 13
##  6 22360     GLASS JAR ENGLISH CONFECTIONERY                13
##  7 22487     WHITE WOOD GARDEN PLANT LADDER                 13
##  8 22626     BLACK KITCHEN SCALES                           13
##  9 21843     RED RETROSPOT CAKE STAND                       12
## 10 21905     MORE BUTTER METAL SIGN                         12
## # i 2,726 more rows
```

```r
# Visualise monthly revenue trend
ggplot(monthly_revenue, aes(x = YearMonth, y = TotalRevenue, group = 1)) +
  geom_line(color = "steelblue") +
  geom_point(color = "darkblue") +
  labs(
    title = "Monthly Revenue Trend",
    x = "Month",
    y = "Total Revenue"
  )
```



Interpretation

Monthly revenue time-series visualisation demonstrates the trend in the number of sales made per month. Going by the graph, one can note that the revenue levels change considerably with time, and there are several maxes and declines in sales over the months. There are also indicators of high sales of about 100,000 units of revenue in several months, which will provide a spike in revenue of the company, indicating high demand.

These spikes indicate that there are some months where the sales activity is high than others. Such growth in revenues can be linked to seasonal retailing needs, promotion efforts or holiday shopping. The retail datasets are usually characterized by more spending rates in certain seasons like at the end of the year or during festivals time when customers buy more items.

Meanwhile, the graph also depicts low revenue times when the sales decrease considerably than during the months of the peaks. These changes are an indication that demand does not remain constant all year round and could be affected by seasonal buying behaviour, availability of the products, or the customer preferences.

The performance analysis of the products over months also points out the list of several products that constantly feature in the list of the low-selling products. An example is the product (StockCode: M) known as Manual, which was seen to be low selling in 35 months thus the most underperforming product in the dataset. Other items like ENAMEL BREAD BIN CREAM, 3 TIER CAKE TIN GREEN and CREAM, and RED RETROSPOT Butter Dishe are also featured in many of the least items sold in several months.

The reason why these products keep on reoccurring as low-selling items can be interpreted to imply that they always have low demands as compared to other products. In business terms, it is significant to know such products since this can result in overstocking, higher storage expenses, or even poor utilization of shelf space. Organizations might be forced to rethink their pricing, promotional activities or products of these items.

In general, the time analysis shows that the sales are not constant throughout the years with periods of high revenue, which might suggest the presence of seasonal tendencies in the customer buying behaviour. Also, the analysis can find a set of products that does not do well on many months, which can be a valuable source of information to make inventory decisions and product portfolio.

I confirm that: - I understand all code submitted. - I can explain my solution if required.